PAVEAISTRUCT™ | PAVEMENT DESIGN & VOLUMETRIC CALCULATION NOTE

DOC. REF: PVMT-CAD-4370

Mechanistic-Empirical Pavement Analysis Suite • IRC:37-2018 / AASHTO Framework

GENERATED: Oct 13, 2026 10:59 PM

Active Section: NH-27/CORR-02/CH:48+200-49+200 • Model Kernel: Multi-Output RF v2.4

RELIABILITY: 90% (Expressway Standard)

REGULATORY COMPLIANCE ASSESSMENT: **IRC:37 COMPLIANT (PASS)** Design Traffic: 50.0 MSA | Carriageway Area: 7,000 m²

1. STRUCTURAL LAYER ARCHITECTURE & INPUT PROPERTIES

Pavement Component	Dimension / Value	Mechanical Properties	Specification Standard
Design Traffic Loading	50.0 MSA	Standard 80 kN Dual-Tire Tandem Axle	IRC:37-2018 Spectrum (90% Rel)
Layer 1: Bituminous Course	150 mm	Dynamic Modulus E• = 3,0	BC Grade-1 + DBM (VG-40 Bitumen @ 35°C)
Layer 2: Unbound Granular Base	250 mm	Equivalent Modulus E, = 2	Wet Mix Macadam (WMM) + GSB Grade-I
Layer 3: Subgrade Soil Medium	Semi-Infinite	Resilient Modulus Ef = 50	Design CBR = 5% (MR = 17.6 × CBR^0.64)
Project Corridor Geometry	Length: 1.0 km	Pavement Width: 7.0 m	Footprint: 7,000 m² (2.5% Camber)

2. MECHANISTIC STRAIN RESPONSES & FAILURE CRITERIA

Mechanic Response Parameter	Surrogate Value	Permissible Limit	Safety Margin	Cumulative Damage	Assessment
Critical Tensile Stra	178.4 με	"d 240.0 με	+25.7% (Margin)	Damage DF = 0.59	**PASS (Optimal)**
Critical Compressive	442.8 με	"d 620.0 με	+28.6% (Margin)	Damage DR = 0.45	**PASS (Safe)**
Fatigue Repetition Capacity (Nf)	1.34e+41	1.00e+35 Min Threshold	Nf = 84.20 Million Repetitions	IRC:37-2018 Transfer Eq	**VERIFIED**
Rutting Capacity (Nr)	1.70e-16	1.00e-14 Max Threshold	Nr = 112.00 Million Repetitions	AASHTO Subgrade Rut Eq	**VERIFIED**

3. BILL OF QUANTITIES (BoQ) & VOLUMETRIC SCHEDULE

Material Layer Description	Thickness	Volume (m³)	Compacted Density	Net Mass (MT)	Logistics (20 MT Dumpers)
Bituminous Courses (BC + DBM)	150 mm	1,050.00 m³	2.40 t/m³	2,520.00 MT	"H 126 Dumpers (20 MT)
Granular Base & Subbase (WMM + GSB)	250 mm	1,750.00 m³	2.10 t/m³	3,675.00 MT	"H 184 Dumpers (20 MT)
TOTAL PAVEMENT STRUCTURE	400 mm	2,800.00 m³	Composite	6,195.00 MT	"H 310 Total Fleet Trips

ENGINEERING CERTIFICATION & PROFESSIONAL VERIFICATION

I hereby certify that the mechanistic responses and material takeoff scheduled herein comply with the specifications prescribed in IRC:37-2018 utilizing the verified Multi-Output Continuum Surrogate Kernel.

IRC:37-2018 CERTIFIED

Designed By: Pavement Design Engineer

Approved By: Lead Highway Consultant